

Uncertainty Quantification in Machine Learning

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Contents

1	Assignment 1 Faite le 7 février 2024 date limite le 8 février 2024	2
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Chapter 1

Assignment 1 Faite le 7 février 2024 date limite le 8 février 2024

Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$. Construct a score function $s(x, y)$ that would yield intervals of the form $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$ for some $\gamma > 0$.

Solution Put

$$s(x, y) = \begin{cases} 2|\hat{f}(x) - y| & y < \hat{f}(x) \\ |\hat{f}(x) - y| & y \geq \hat{f}(x). \end{cases}$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $\gamma = \frac{\hat{q}}{2}$. ////

Exercise 2

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}^d$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ whose coordinates are $\hat{f}(x) = (\hat{f}_1(x), \dots, \hat{f}_d(x))$. Construct a score function $s(x, y)$ that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^d [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a $c > 0$.

Solution Put

$$s(x, y) = \max_{j=1, \dots, d} |\hat{f}_j(x) - y_j|.$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $c = \hat{q}$. ////

Exercise 3

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$, two given models $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$ for $i = 1, 2$, and the score function

$$s(x, y) = \min_{i=1,2} |y - \hat{f}_i(x)|.$$

Derive an explicit expression for $\mathcal{C}(x)$ in terms of $\hat{f}_1(x)$, $\hat{f}_2(x)$, and the quantile $\hat{q}$ of the scores on the validation set. What is the shape of the prediction when $\hat{q}$ is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of Y given $X = x$ is bimodal.

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ///

Exercise 4

Fix $n > 2$. Construct a random vector $(X_1, \dots, X_n)$ defined on some probability space such that (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable.

Solution Let (X, Y) be a jointly normal random vector. It is exchangeable if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\text{Var}(X) = \text{Var}(Y)$. With this in mind, let $(X_1, \dots, X_n)$ be a jointly normal random vector such that $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ and $\text{Var}(X_i) = \text{Var}(X_j)$ for all $i \neq j$ but $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$. Then (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable. For a concrete example, say $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = 1$ for all i , $\text{Cov}(X_1, X_2) = -1$, and all other covariances are zero. ///

Exercise 5

Fix $n \geq 1$, $\tau \in (0, 1)$. For any vector $v \in \mathbb{R}^n$, define $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$, where $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$ are the order statistics of v . Under what condition is it true that, for all $v \in \mathbb{R}^n$, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$?

Solution The order statistics of $-v$ are given by $(-v)_{(i)} = -v_{(n-i+1)}$ for $i = 1, \dots, n$. Therefore, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$ holds for a vector $v \in \mathbb{R}^n$ if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n-\lceil \tau n \rceil+1)}) = v_{(n-\lceil \tau n \rceil+1)}.$$

To make this work for all $v \in \mathbb{R}^n$, the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when $\tau = 1/2$ and n is odd. ///

Exercise 6

For $z \in \mathbb{R}^n$, write $\hat{F}_z(v) = \frac{1}{n} \sum_{i=1}^n \chi_{\{z_i \leq v\}}$ for the empirical cumulative distribution function of z with $\hat{F}_z(-\infty) = 0$ and $\hat{F}_z(+\infty) = 1$. Write

$$\text{Quantile}(z; \tau) = \inf \left\{ v \in \mathbb{R} \mid \hat{F}_z(v) \geq \tau \right\}$$

for the empirical τ -quantile of z for any $\tau \in [0, 1]$. Consequently, $\text{Quantile}(z; 0) = -\infty$.

Prove that, for any $z \in \mathbb{R}^n$, the following statements are true:

1. $\hat{F}_z(v) = \sup \{ \tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v \}$ for all $v \in \mathbb{R}$.
2. $\text{Quantile}(z; \hat{F}_z(v)) \leq v$ for all $v \in \mathbb{R}$.
3. $\hat{F}_z(\text{Quantile}(z; \tau)) \geq \tau$ for all $\tau \in [0, 1]$.
4. If z has no repeated values, then $\hat{F}_z(\text{Quantile}(z; \tau)) = \frac{\lfloor n\tau \rfloor}{n}$ for all $\tau \in [0, 1]$.

Solution Let $z \in \mathbb{R}^n$ be given and fixed. Let us introduce a notation S_τ for the set of all $v \in \mathbb{R}$ such that $\hat{F}_z(v) \geq \tau$. Then, by definition, $\text{Quantile}(z; \tau) = \inf S_\tau$. All four statements depend on the following lemma:

Lemma 1. If $v \in \mathbb{R}$ and $\tau \in [0, 1]$, then $\text{Quantile}(z; \tau) \leq v$ if and only if $\tau \leq \hat{F}_z(v)$.

Proof. If $\tau \leq \hat{F}_z(v)$, then $v \in S_\tau$ and $v \geq \inf S_\tau = \text{Quantile}(z; \tau)$.

Let us focus on the converse. Suppose that $\text{Quantile}(z; \tau) \leq v$. For simplicity, put $q = \text{Quantile}(z; \tau)$. If $\tau = 0$, then $\hat{F}_z(v) \geq 0 = \tau$ holds trivially. Assume from now on that $\tau > 0$. Then q is finite number. Choose $\delta > 0$ small enough such that no points z_i lies between q and $q + \delta$, which is always doable because z has only finitely many points. Since q is the infimum, choose $s \in S_\tau$ such that $q \leq s < q + \delta$. Then, we arrive at

$$\hat{F}_z(v) \geq \hat{F}_z(q) = \hat{F}_z(s) \geq \tau$$

by applying in the first inequality the monotonicity of $\hat{F}_z$ and in the second equality the fact that no points z_i lies between q and $q + \delta$. The lemma is proved. $\square$

We now prove the four statements. To prove the first statement, let $v \in \mathbb{R}$ be given and we have, by the lemma:

$$\{ \tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v \} = \left\{ \tau \in [0, 1] \mid \tau \leq \hat{F}_z(v) \right\} = [0, \hat{F}_z(v)]$$

and the supremum of the interval is just $\hat{F}_z(v)$. The second and the third statements are already included in the lemma. The fourth statement is trivial given the special assumption that z has no repeated values.////

Exercise 7

exercise 3.1

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{ y \in \mathcal{Y} \mid s(x, y) \leq \hat{q} \} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

Exercise 8

exercise 3.6

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

Exercise 9

exercise 3.7

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////